

YEAR 7 MATHEMATICS

Chapter 3

Whole numbers

MIXED PRACTICE WORKSHEETS

3

How to use this booklet

- Each lesson is **one double-sided sheet**. Write your name, class and date at the top.
- Then work down the **columns**, left to right. They get harder as you go: **Easy** **Easy** **Medium** on page 1, then **Medium** **Challenging** **Challenging** on page 2.
- Most questions practise **today's skill**. A question marked **from 1.03** comes from an earlier lesson, to keep it fresh. Stuck? Look back at that lesson.
- Each question tells you what to do: read it carefully. Show your working on the lines. The first question with working in each Easy column shows you how to set it out.

WE DO

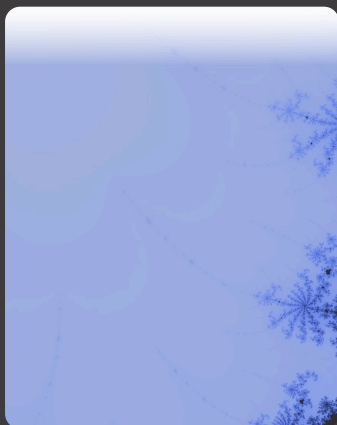
The grey box at the top. Your teacher shows the key idea with two or three examples. **Copy every step** into the space.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
3.01	Rounding and estimation	3		☐
3.02	Multiplication	5		☐
3.03	Division	7		☐
3.04	Divisibility tests	9		☐
3.05	Long division	11		☐
3.06	Roman numerals	13		☐
3.07	Index notation and powers	15		☐
3.08	Square roots and cube roots	17		☐
3.09	Prime and composite numbers	19		☐
3.10	Prime factors	21		☐
3.11	Highest common factor	23		☐
3.12	Lowest common multiple	25		☐

In this chapter you will

- round and estimate
- multiply and divide whole numbers
- use divisibility tests
- use long division and Roman numerals
- use index notation, square roots and cube roots
- find prime numbers and prime factors
- find the HCF and LCM of two numbers



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the tentacled edge of a small copy of the whole set, out on the antenna near -1.77 . Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

To round, look at the digit to the right of the place you are rounding to: 5 or more, round up; 4 or less, round down. To estimate, round each number to its first digit, then work it out.

E1 Round 6 482 to the nearest hundred.

E2 Estimate 389×52 .

E3 27 students each pay \$48. Estimate the total.

Easy

Questions 1–9

1 Round 1 531 to the nearest hundred.

2 Round 9 106 to its first digit.

3 Is 2 240 closer to 2 000 or 3 000?

4 Round 796 to the nearest ten.

5 Round 708 to its first digit.

6 Is 2 190 closer to 2 000 or 3 000?

7 Round 274 to the nearest ten.

8 Round 879 to its first digit.

9 $3 - (-3)$

from 1.05

Easy

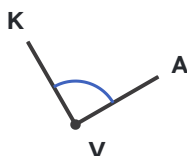
Questions 10–17

10 Round 9 007 to its first digit.

11 Is 2 561 closer to 2 000 or 3 000?

12 Round 551 to the nearest ten.

13 Name the marked angle with three letters.



from 2.01

14 Estimate 30×14 .

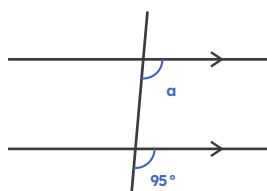
$\approx \text{---} \times \text{---}$

$\approx \text{---}$

15 Round 1 303 to its first digit.

16 Is 2 784 closer to 2 000 or 3 000?

17 The lines are parallel. Find a .



from 2.10

Medium

Questions 18–25

18 Estimate $48 + 72$.

19 Estimate $1\,946 \div 73$.

20 Start at 4. Move 9 left. Where do you land?

from 1.02

21 Round 88 200 to the nearest thousand.

22 Estimate $8\,390 - 3\,206$.

23 Order from largest to smallest: -10 , -25 , -2 , 9

from 1.03

24 Estimate $43 + 64$.

25 Estimate $5\,842 \div 38$.

Medium

Questions 26–36

26 Estimate $7\,990 \div 47$.

27 Round 95 655 to the nearest thousand.

28 What is the opposite of 2?

from 1.01

29 Estimate $6\,957 - 3\,410$.

30 The highest temperature was 18°C . The lowest was -14°C . What is the difference?

from 1.10

31 Estimate 77×74 .

32 32 students each pay \$14 for an excursion. Estimate the total cost.

33 $5 \times (-3) - 3$

from 1.08

34 Estimate $5\,116 \div 78$.

35 $14 + (-31)$

from 1.04

36 Round 20 961 to the nearest thousand.

Challenging

Questions 37–45

37 26 students each pay \$37 for an excursion. Estimate the total cost.

38 783 people travel in 25 buses. About how many are on each bus?

39 To draw a reflex angle of 315° , what angle do you draw first?

from 2.02

40 Ruby says $661 \times 16 = 105\,760$. Use an estimate to check. Is it reasonable?

41 Co-interior angles are $2x^{\circ}$ and 142° . Find x so that $AB \parallel CD$.

from 2.11

42 What is the smallest whole number that rounds to 300 (to the nearest hundred)?

43 24 students each pay \$16 for an excursion. Estimate the total cost.

44 $-40 \div (-8)$

from 1.07

45 822 people travel in 22 buses. About how many are on each bus?

Challenging

Questions 46–54

46 732 people travel in 33 buses. About how many are on each bus?

47 What is the largest whole number that rounds to 400 (to the nearest hundred)?

48 The lines are parallel. An angle is 58° . What is its corresponding angle?

from 2.07

49 40 students each pay \$31 for an excursion. Estimate the total cost.

50 $-3 \times (-2) \times 3$

from 1.06

51 Estimate $8\,174 - 4\,673$.

52 359 people travel in 31 buses. About how many are on each bus?

53 Find the supplement of 69° .

from 2.04

54 What is the largest whole number that rounds to 800 (to the nearest hundred)?

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Split a number to multiply in your head: $34 \times 6 = 30 \times 6 + 4 \times 6$. For bigger numbers, multiply by each part of the second number, then add.

Watch your teacher.
Copy every step.

E1 34×6

E2 56×300

E3 243×17

Easy

Questions 1–12

1 10×10

2 75×10

3 40×9

4 17×5

= ____ + ____

= ____

5 8×11

6 $32 \times 1\,000$

7 80×8

8 37×3

9 Round 41 to its first digit.

from 3.01

10 9×8

11 $72 \times 1\,000$

12 70×4

Easy

Questions 13–22

13 77×100

14 60×3

15 31×3

= ____ + ____

= ____

16 For $AB \parallel CD$, what must be true of corresponding angles?

from 2.11

17 12×10

18 71×5

19 37×10

20 Is 2 334 closer to 2 000 or 3 000?

from 3.01

21 40×5

22 48×9

Medium

Questions 23–29

23 76×7

24 234×5

25 Estimate $7\,808 - 4\,648$.

from 3.01

26 80×900

27 A box holds 12 stickers. How many stickers are in 42 boxes?

28 The lines are parallel. An angle is 119° . What is its corresponding angle?

from 2.07

29 95×6

Medium Questions 30–40	Challenging Questions 41–50	Challenging Questions 51–60
30 326×9 _____ _____	41 306×31 _____ _____	51 Use $99 = 100 - 1$ to work out 99×55. _____ _____
31 60×800 _____ _____	42 Use $99 = 100 - 1$ to work out 99×50. _____ _____	52 A hall has 17 rows of 17 seats. Tickets cost \$14. How much money is taken if every seat is sold? _____ _____
32 Round 23 to its first digit. from 3.01	43 Estimate $6\ 830 - 1\ 923$. from 3.01	53 Use a calculator: $(-20)^2$ from 1.09
33 A box holds 17 cards. How many cards are in 25 boxes? _____ _____	44 A hall has 18 rows of 29 seats. Tickets cost \$15. How much money is taken if every seat is sold? _____ _____	54 Find the missing digit: $\square \times 4 = 356$ _____ _____
34 Estimate $9\ 216 \div 29$. from 3.01	45 $8 + (-3)$ from 1.04	55 Round 48 to its first digit. from 3.01
35 78×6 _____ _____	46 Find the missing digit: $\square \times 3 = 84$ _____ _____	56 426×23 _____ _____
36 418×22 _____ _____	47 241×15 _____ _____	57 A box holds 20 cards. How many cards are in 25 boxes? _____ _____
37 Order from largest to smallest: -15, -9, -21, 2 from 1.03	48 Is 3 303 closer to 3 000 or 4 000? from 3.01	58 Estimate $5\ 116 - 3\ 051$. from 3.01
38 393×7 _____ _____	49 Use $99 = 100 - 1$ to work out 99×30. _____ _____	59 Use $99 = 100 - 1$ to work out 99×52. _____ _____
39 Round 375 to the nearest ten. from 3.01	50 Estimate $8\ 966 \div 18$. from 3.01	60 Are the lines in train tracks parallel, perpendicular or neither? from 2.06
40 40×400 _____ _____		

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Division facts come from the times tables. Split a number into parts you can divide easily. Short division: divide each place from the left and carry the remainder.

Watch your teacher.
Copy every step.

E1 $72 \div 8$

E2 $4\,500 \div 100$

E3 $852 \div 4$

Easy

Questions 1–13

1 $21 \div 3$

2 $5\,100 \div 100$

3 How many 5s are in 60?

4 $120 \div 12$

5 $330 \div 10$

6 How many 5s are in 45?

7 $33 \div 11$

8 $250 \div 10$

9 $15 \times 1\,000$

from 3.02

10 How many 8s are in 64?

11 $99 \div 11$

12 $3\,600 \div 100$

13 How many 9s are in 81?

Easy

Questions 14–22

14 $100 \div 10$

15 How many 5s are in 40?

16 $20 \div 5$

17 Are the lines in train tracks parallel, perpendicular or neither?

from 2.06

18 $328 \div 8$

= ____ + ____

= ____

19 $110 \div 10$

20 How many 7s are in 56?

21 Round 3 502 to its first digit.

from 3.01

22 $77 \div 7$

Medium

Questions 23–30

23 $88 \div 8$

24 $1\,141 \div 7$

25 80×500

from 3.02

26 $144 \div 5$. Give the remainder.

27 $576 \div 9$

28 The lines are parallel. What is the alternate angle to 63° ?

from 2.08

29 $1\,057 \div 7$

30 $258 \div 5$. Give the remainder.

Medium

Questions 31–40

31 $1\,296 \div 6$

32 $206 \div 6$. Give the remainder.

33 Is 7 806 closer to 7 000 or 8 000?

from 3.01

34 $198 \div 6$

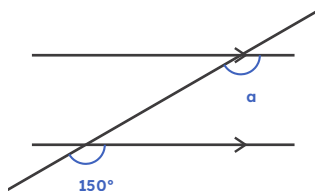
35 A box holds 16 cards. How many cards are in 28 boxes?

from 3.02

36 271 lollies are shared by 9 people. How many does each get, and how many are left over?

37 $1\,092 \div 6$

38 Find a . Give a reason.



from 2.07

39 $437 \div 3$. Give the remainder.

40 Round 30 683 to the nearest thousand.

from 3.01

Challenging

Questions 41–50

41 450 lollies are shared by 7 people. How many does each get, and how many are left over?

42 70 people need a ride. Each van holds 9. How many vans are needed?

43 74×6

from 3.02

44 Find the missing number:
 $\square \div 8 = 25$

45 Start at -3 . Move 6 right. Where do you land?

from 1.02

46 $7\,182 \div 7$

47 165 lollies are shared by 6 people. How many does each get, and how many are left over?

48 Round 9 205 to its first digit.

from 3.01

49 97 people need a ride. Each van holds 7. How many vans are needed?

50 123×5

from 3.02

Challenging

Questions 51–59

51 73 people need a ride. Each van holds 6. How many vans are needed?

52 Find the missing number:
 $\square \div 5 = 41$

53 Find the angle between the clock hands at 8:00. Classify it.

from 2.03

54 $19\,728 \div 8$

55 Is 6 271 closer to 6 000 or 7 000?

from 3.01

56 238 lollies are shared by 5 people. How many does each get, and how many are left over?

57 $499 \div 7$. Give the remainder.

58 60×700

from 3.02

59 79 people need a ride. Each van holds 6. How many vans are needed?

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Use a test, not division. 2: even. 5: ends in 0 or 5. 10: ends in 0. 3 and 9: check the digit sum. 4: check the last two digits. 6: divisible by 2 and 3. 8: check the last three digits.

Watch your teacher.
Copy every step.

E1 Is 4 527 divisible by 3?

E2 Is 7 316 divisible by 4?

E3 Is 5 742 divisible by 6?

Easy

Questions 1–9

1 Is 5 815 divisible by 2?

2 Find the digit sum of 8 450. Is 8 450 divisible by 3?

3 Which number is divisible by 5: 1 333, 395, 999, 7 663?

4 Is 8 221 divisible by 2?

5 Find the digit sum of 5 069. Is 5 069 divisible by 3?

6 Which number is divisible by 5: 3 453, 1 261, 895, 1 353?

7 Is 3 128 divisible by 2?

8 Find the digit sum of 5 837. Is 5 837 divisible by 3?

9 $64\,000 \div 1\,000$

from 3.03

Easy

Questions 10–18

10 Find the digit sum of 7 135. Is 7 135 divisible by 3?

11 Which number is divisible by 5: 875, 319, 737, 7 213?

12 Is 5 979 divisible by 5?

13 The lines are parallel. What is the co-interior angle to 31° ?

from 2.09

14 Is 5 857 divisible by 9?

digit sum: ____

yes / no: _____

15 Find the digit sum of 4 830. Is 4 830 divisible by 3?

16 Which number is divisible by 5: 1 549, 1 487, 5 533, 455?

17 34×10

from 3.02

18 Is 8 563 divisible by 2?

Medium

Questions 19–26

19 Is 8 318 divisible by 9?

20 Is 8 624 divisible by 4?

21 Estimate $7\,979 \div 72$.

from 3.01

22 Is 9 828 divisible by 6?

23 Is 47 804 divisible by 8?

24 $-2 \times (-9)$

from 1.06

25 Is 5 993 divisible by 9?

26 Is 2 855 divisible by 4?

Medium

Questions 27–37

27 Is 3 264 divisible by 4?

28 Is 6 893 divisible by 6?

29 How many 8s are in 48?

from 3.03

30 Is 92 794 divisible by 8?

31 20×200

from 3.02

32 Is 9 777 divisible by 9?

33 $516\square$ is divisible by 3. Find every digit that could go in the box.

34 Which integer is halfway between -4 and 8 ?

from 1.02

35 Is 7 447 divisible by 4?

36 Is 2 726 closer to 2 000 or 3 000?

from 3.01

37 Is 6 591 divisible by 2?

Challenging

Questions 38–46

38 $164\square$ is divisible by 9. Find every digit that could go in the box.

39 Which of 2, 3, 4, 5, 6, 8, 9 and 10 divide 7 356?

40 $432 \div 8$

from 3.03

41 Leo says 1 310 is divisible by 4 because it ends in 0. Is that right? Explain.

42 $-6 - (-6)$

from 1.05

43 What is the smallest four-digit number that is divisible by 4?

44 $261\square$ is divisible by 3. Find every digit that could go in the box.

45 42×7

from 3.02

46 Which of 2, 3, 4, 5, 6, 8, 9 and 10 divide 3 720?

Challenging

Questions 47–56

47 Which of 2, 3, 4, 5, 6, 8, 9 and 10 divide 9 396?

48 Chloe says 9 858 is divisible by 4 because it ends in 8. Is that right? Explain.

49 Find c .



from 2.05

50 What is the smallest four-digit number that is divisible by 8?

51 $39\,000 \div 1\,000$

from 3.03

52 $511\square$ is divisible by 9. Find every digit that could go in the box.

53 Is 99 163 divisible by 8?

54 81×5

from 3.02

55 Which of 2, 3, 4, 5, 6, 8, 9 and 10 divide 8 328?

56 Find the supplement of 25° .

from 2.04

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

Long division: divide, multiply, subtract, bring down the next digit. Repeat until there are no digits left. What is left at the end is the remainder.

E1 $672 \div 16$

E2 $851 \div 23$

E3 500 people go in buses that hold 45. How many buses?

Easy

Questions 1–9

1 23×5

2 How many times does 20 go into 80?

3 How many times does 21 go into 46? What is left over?

4 18×6

5 How many times does 15 go into 60?

6 How many times does 17 go into 129? What is left over?

7 18×7

8 How many times does 15 go into 45?

9 $6\,200 \div 100$

from 3.03

Easy

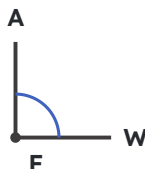
Questions 10–18

10 How many times does 24 go into 120?

11 How many times does 17 go into 82? What is left over?

12 18×3

13 Name the marked angle with three letters.



from 2.01

14 $684 \div 12$

$68 \div 12 = \underline{\hspace{2cm}}$

bring down: $\underline{\hspace{1cm}} \div \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

$= \underline{\hspace{1cm}}$

15 How many times does 24 go into 192?

16 How many times does 15 go into 124? What is left over?

17 Find the digit sum of 5 132. Is 5 132 divisible by 3?

from 3.04

18 14×9

Medium

Questions 19–26

19 $560 \div 14$

20 $3\,920 \div 16$

21 Estimate $8\,823 \div 89$.

from 3.01

22 $423 \div 17$. Give the remainder.

23 $779 \div 19$

24 321×7

from 3.02

25 $5\,440 \div 34$

26 $998 \div 24$. Give the remainder.

Medium

Questions 27–37

27 $6\,765 \div 33$

28 $694 \div 24$. Give the remainder.

29 The lines are parallel. What is the co-interior angle to 50° ?

from 2.09

30 $528 \div 22$

31 Is 6 850 closer to 6 000 or 7 000?

from 3.01

32 758 students are put into teams of 21. How many full teams are there? How many students are left over?

33 $2\,460 \div 12$

34 How many 7s are in 49?

from 3.03

35 $514 \div 15$. Give the remainder.

36 Is 3 703 divisible by 6?

from 3.04

37 $768 \div 12$

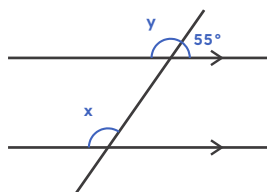
Challenging

Questions 38–45

38 781 students are put into teams of 26. How many full teams are there? How many students are left over?

39 671 people go to a game in buses that hold 51. How many buses are needed?

40 Find y , then x . Give reasons.



from 2.10

41 A club shares \$4 125 equally between 15 members. How much does each get?

42 90×4

from 3.02

43 When a number is divided by 19, the answer is 32 remainder 18. What is the number?

44 1 083 students are put into teams of 38. How many full teams are there? How many students are left over?

45 Estimate $5\,592 - 1\,741$.

from 3.01

Challenging

Questions 46–54

46 778 people go to a game in buses that hold 42. How many buses are needed?

47 A club shares \$2 717 equally between 13 members. How much does each get?

48 Chloe has \$66 in the bank and spends \$103. What is the balance?

from 1.10

49 When a number is divided by 14, the answer is 29 remainder 8. What is the number?

50 Round 8 702 to its first digit.

from 3.01

51 937 students are put into teams of 25. How many full teams are there? How many students are left over?

52 $546 \div 24$. Give the remainder.

53 Is 6 325 divisible by 2?

from 3.04

54 986 people go to a game in buses that hold 45. How many buses are needed?

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

I = 1, V = 5, X = 10, L = 50, C = 100, D = 500, M = 1000. A smaller letter after a larger one adds (VI = 6); a smaller letter before a larger one subtracts (IV = 4, XC = 90).

Watch your teacher.
Copy every step.

E1 Write XLVII as a number.

E2 Write 94 in Roman numerals.

E3 Write 2026 in Roman numerals.

Easy

Questions 1–9

1 Write LIII as a number.

2 Write 36 in Roman numerals.

3 What number is V?

4 Write LV as a number.

5 Write 61 in Roman numerals.

6 What number is X?

7 Write XXV as a number.

8 Write 26 in Roman numerals.

9 How many times does 20 go into 80?

from 3.05

Easy

Questions 10–18

10 Write 67 in Roman numerals.

11 What number is L?

12 Write LXV as a number.

13 $-1 - (-3)$

from 1.05

14 Write 264 in Roman numerals.

___ + ___ + ___

15 Write 93 in Roman numerals.

16 What number is M?

17 Find the digit sum of 9 044.
Is 9 044 divisible by 3?

from 3.04

18 Write XLI as a number.

Medium

Questions 19–26

19 Write 263 in Roman numerals.

20 Write CDLXV as a number.

21 Estimate $4\,965 \div 47$.

from 3.01

22 Which is larger: XLVIII or LXXXVI?

23 Write 719 in Roman numerals.

24 $1\,359 \div 9$

from 3.03

25 Write CMXCVII as a number.

26 Which is larger: XLV or XXVIII?

Medium

Questions 27–37

- 27 Write CXXI as a number.
-
-
- 28 Which is larger: XCI or LVII?
-
-
- 29 Use a calculator: -34×31
- from 1.09
- 30 Write 787 in Roman numerals.
-
-
- 31 73×10
- from 3.02
- 32 Write the year 1975 in Roman numerals.
-
-
- 33 Write CMIV as a number.
-
-
- 34 How many times does 23 go into 172? What is left over?
- from 3.05
- 35 Which is larger: LVI or LXIII?
-
-
- 36 Is 6 152 divisible by 6?
- from 3.04
- 37 Write 330 in Roman numerals.
-
-

Challenging

Questions 38–47

- 38 Write the year 1894 in Roman numerals.
-
-
- 39 Work out LVIII + XV. Write the answer in Roman numerals.
-
-
- 40 Find the angle between the clock hands at 4:00. Classify it.
- from 2.03
- 41 Mason writes 95 as VC. Is that right? Explain.
-
-
- 42 How many 7s are in 42?
- from 3.03
- 43 Write the year 1961 in Roman numerals.
-
-
- 44 Work out XIX + XLV. Write the answer in Roman numerals.
-
-
- 45 Round 74 776 to the nearest thousand.
- from 3.01
- 46 Zac writes 999 as IM. Is that right? Explain.
-
-
- 47 $702 \div 18$
- from 3.05

Challenging

Questions 48–57

- 48 Work out LIII + XIV. Write the answer in Roman numerals.
-
-
- 49 Mason writes 99 as IC. Is that right? Explain.
-
-
- 50 Which is larger: 4 or 5?
- from 1.03
- 51 Write the year 1810 in Roman numerals.
-
-
- 52 40×4
- from 3.02
- 53 Which is larger: XCVIII or LXX?
-
-
- 54 Work out LVIII + XXXV. Write the answer in Roman numerals.
-
-
- 55 Is 7 399 divisible by 5?
- from 3.04
- 56 Zac writes 49 as IL. Is that right? Explain.
-
-
- 57 $136 \div 8$
- from 3.03

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

A power shows repeated multiplication: $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$. The base is 2 and the index (the power) is 5.

Watch your teacher.
Copy every step.

E1 Write $3 \times 3 \times 3 \times 3$ in index form.

E2 Evaluate 2^5 .

E3 Evaluate $4^2 + 2^3$.

Easy

Questions 1–9

1 Write 9×9 in index form.

2 Evaluate 3^2 .

3 Evaluate 3^3 .

$= ___ \times ___ \times ___$

$= ___$

4 Write $4 \times 4 \times 4$ in index form.

5 Evaluate 2^2 .

6 Evaluate 10^4 .

7 Write $4 \times 4 \times 4 \times 4 \times 4$ in index form.

8 Evaluate 12^2 .

9 $19 \times 1\,000$

from 3.02

Easy

Questions 10–18

10 Evaluate 11^2 .

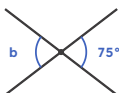
11 Evaluate 10^3 .

$= ___ \times ___ \times ___$

$= ___$

12 Write $8 \times 8 \times 8$ in index form.

13 Find b .



from 2.05

14 Write 4^5 as a product. Name the base and the index.

15 Evaluate 7^2 .

16 Evaluate 4^3 .

17 Write 91 in Roman numerals.

from 3.06

18 Write $2 \times 2 \times 2 \times 2$ in index form.

Medium

Questions 19–26

19 Write 7^5 as a product. Name the base and the index.

20 Evaluate 5^4 .

21 Estimate $7\,210 \div 21$.

from 3.01

22 Which is larger: 2^3 or 3^2 ?

23 Evaluate $4^2 + 3^3$.

24 $3\,332 \div 28$

from 3.05

25 Write 3^3 as a product. Name the base and the index.

26 Evaluate 2^8 .

Medium

Questions 27–37

27 Evaluate 7^3 .

28 Which is larger: 4^5 or 5^4 ?

29 $1 - (-6)$

from 1.05

30 Evaluate $3^2 + 2^3$.

31 $4\,100 \div 100$

from 3.03

32 Write 8^6 as a product. Name the base and the index.

33 Write $2 \times 2 \times 5$ in index form, then evaluate it.

34 Find the digit sum of 3 603. Is 3 603 divisible by 3?

from 3.04

35 Evaluate 6^3 .

36 30×800

from 3.02

37 Which is larger: 2^5 or 5^2 ?

Challenging

Questions 38–47

38 Write $2 \times 2 \times 5 \times 5$ in index form, then evaluate it.

39 Find the index: $3^n = 81$

40 $5 \times (-5) \times (-1)$

from 1.06

41 Write 1 000 000 as a power of 10.

42 What number is L?

from 3.06

43 Sienna says $3^2 = 6$. Is that right? Explain.

44 Write $2 \times 2 \times 2 \times 2 \times 5 \times 5$ in index form, then evaluate it.

45 Round 95 819 to the nearest thousand.

from 3.01

46 Find the index: $4^n = 64$

47 $553 \div 23$. Give the remainder.

from 3.05

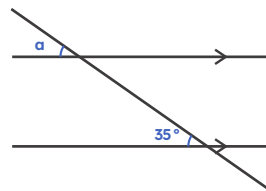
Challenging

Questions 48–57

48 Find the index: $3^n = 243$

49 Write 1 000 as a power of 10.

50 The lines are parallel. Find a .



from 2.10

51 Kai says $6^2 = 12$. Is that right? Explain.

52 How many 11s are in 44?

from 3.03

53 Write $2 \times 2 \times 2 \times 5$ in index form, then evaluate it.

54 Evaluate $3^2 + 3^3$.

55 Which number is divisible by 5: 737, 975, 1 299, 7 023?

from 3.04

56 Find the index: $2^n = 32$

57 A box holds 7 cards. How many cards are in 16 boxes?

from 3.02

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

A square root undoes squaring: $\sqrt{49} = 7$ because $7^2 = 49$. A cube root undoes cubing: $\sqrt[3]{27} = 3$ because $3^3 = 27$.

Watch your teacher.
Copy every step.

E1 $\sqrt{64}$

E2 $\sqrt[3]{125}$

E3 Between which two whole numbers is $\sqrt{40}$?

Easy

Questions 1–13

1 $\sqrt{25}$

2 $\sqrt[3]{64}$

3 What number squared is 100?

4 $\sqrt{49}$

5 $\sqrt[3]{8}$

6 What number squared is 49?

7 $\sqrt{121}$

8 $\sqrt[3]{27}$

9 Evaluate 4^2 .

from 3.07

10 What number squared is 9?

11 $\sqrt{16}$

12 $\sqrt[3]{125}$

13 What number squared is 64?

Easy

Questions 14–22

14 $\sqrt[3]{1}$

15 What number squared is 81?

16 $\sqrt{144}$

17 What is the opposite of -29 ?

from 1.01

18 Between which two whole numbers is $\sqrt{101}$?

102 = _____

between ____ and ____

19 What number squared is 16?

20 $\sqrt{9}$

21 Write 13 in Roman numerals.

from 3.06

22 Between which two whole numbers is $\sqrt{95}$?

Medium

Questions 23–30

23 Between which two whole numbers is $\sqrt{8}$?

24 $\sqrt{4} + \sqrt{49}$

25 Estimate $4\,940 \div 79$.

from 3.01

26 $\sqrt[3]{125} \times \sqrt{4}$

27 $\sqrt[3]{512}$

28 $1\,963 \div 13$

from 3.05

29 Between which two whole numbers is $\sqrt{114}$?

30 $\sqrt{49} + \sqrt{25}$

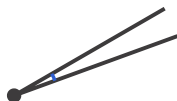
Medium

Questions 31–41

31 $\sqrt{25} + \sqrt{4}$

32 $\sqrt[3]{64} \times \sqrt{16}$

33 Classify the angle.



from 2.03

34 $\sqrt[3]{1\,000}$

35 51×10

from 3.02

36 Between which two whole numbers is $\sqrt{136}$?

37 $\sqrt{(9 + 16)}$

38 Evaluate 2^4 .

from 3.07

39 $\sqrt{4} + \sqrt{4}$

40 Which is larger: LXXVI or LXVI?

from 3.06

41 $\sqrt[3]{125} \times \sqrt{64}$

Challenging

Questions 42–51

42 $\sqrt{(64 + 225)}$

43 A square has an area of 169 cm^2 . How long is each side?

44 $-1 \times (-1) \times 1$

from 1.06

45 A cube has a volume of 8 cm^3 . How long is each edge?

46 How many times does 19 go into 106? What is left over?

from 3.05

47 Is $\sqrt{(16 + 9)}$ the same as $\sqrt{16} + \sqrt{9}$? Show why.

48 $\sqrt{(81 + 144)}$

49 $600 \div 3$

from 3.03

50 A square has an area of 144 cm^2 . How long is each side?

51 Evaluate $6^2 + 3^3$.

from 3.07

Challenging

Questions 52–61

52 A square has an area of 36 cm^2 . How long is each side?

53 A cube has a volume of 125 cm^3 . How long is each edge?

54 The lines are parallel. What is the alternate angle to 57° ?

from 2.08

55 $\sqrt{(36 + 64)}$

56 Find the digit sum of 6 081. Is 6 081 divisible by 3?

from 3.04

57 $\sqrt[3]{216}$

58 A square has an area of 121 cm^2 . How long is each side?

59 Write XLIV as a number.

from 3.06

60 A cube has a volume of 27 cm^3 . How long is each edge?

61 $684 \div 18$

from 3.05

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

A prime number has exactly two factors: 1 and itself. A composite number has more than two factors. 1 is neither prime nor composite.

Watch your teacher.
Copy every step.

E1 List the factors of 18.

E2 Is 51 prime or composite?

E3 List the primes between 20 and 40.

Easy

Questions 1–9

1 Is 48 prime or composite?

2 List the factors of 18.

3 How many factors does 6 have?

4 Is 30 prime or composite?

5 List the factors of 26.

6 How many factors does 27 have?

7 Is 31 prime or composite?

8 List the factors of 10.

9 Find the digit sum of 8 359. Is 8 359 divisible by 3?

from 3.04

Easy

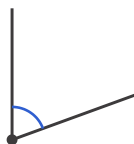
Questions 10–18

10 List the factors of 35.

11 How many factors does 18 have?

12 Is 36 prime or composite?

13 Classify the angle.



from 2.03

14 Is 81 prime or composite? Give a reason.

15 List the factors of 38.

16 How many factors does 8 have?

17 $\sqrt[3]{8}$

from 3.08

18 Is 49 prime or composite?

Medium

Questions 19–26

19 Is 90 prime or composite? Give a reason.

20 List the primes between 40 and 60.

21 Estimate $8\,827 \div 41$.

from 3.01

22 Write 34 as the sum of two primes.

23 Is 76 prime or composite? Give a reason.

24 Evaluate 5^4 .

from 3.07

25 List the primes between 10 and 30.

26 Write 28 as the sum of two primes.

Medium

Questions 27–37

- 27 List the primes between 70 and 90.

- 28 Write 18 as the sum of two primes.

- 29 Start at 4. Move 2 right. Where do you land?

from 1.02

- 30 Is 80 prime or composite? Give a reason.

- 31 12×100

from 3.02

- 32 What is the smallest prime number greater than 35?

- 33 List the primes between 20 and 40.

- 34 Write 79 in Roman numerals.

from 3.06

- 35 Write 38 as the sum of two primes.

- 36 Is 4 648 divisible by 6?

from 3.04

- 37 Is 55 prime or composite? Give a reason.

Challenging

Questions 38–47

- 38 What is the smallest prime number greater than 106?

- 39 Ruby says 91 is prime. Is that right? Explain.

- 40 The temperature went from -6°C to 10°C . How much did it rise?

from 1.05

- 41 Two primes add to 15. What are they?

- 42 What number squared is 81?

from 3.08

- 43 92 has how many factors? List them.

- 44 What is the smallest prime number greater than 107?

- 45 $774 \div 6$

from 3.03

- 46 Jack says 119 is prime. Is that right? Explain.

- 47 Which is larger: 2^3 or 3^2 ?

from 3.07

Challenging

Questions 48–57

- 48 Jack says 51 is prime. Is that right? Explain.

- 49 Which is the only even prime number? Why is it the only one?

- 50 Find the supplement of 169° .

from 2.04

- 51 70 has how many factors? List them.

- 52 How many times does 21 go into 63?

from 3.05

- 53 What is the smallest prime number greater than 39?

- 54 Write 30 as the sum of two primes.

- 55 What number is C?

from 3.06

- 56 Priya says 57 is prime. Is that right? Explain.

- 57 Is 79 173 divisible by 8?

from 3.04

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Break a number into factors until every branch of the factor tree ends in a prime.

Watch your teacher.
Copy every step.

Write the primes in order, using index notation: $60 = 2^2 \times 3 \times 5$.

E1 Write 36 as a product of primes.

E2 Write 90 as a product of primes.

E3 Which number is $2^3 \times 5$?

Easy

Questions 1–9

1 Which number is $3^3 \times 7$?

= ____ × ____

= ____

2 Write $2 \times 2 \times 2 \times 3 \times 3 \times 3$ in index form.

3 Which primes are factors of 50?

4 Which number is $3^3 \times 5$?

5 Write $2 \times 3 \times 3$ in index form.

6 Which primes are factors of 28?

7 Which number is $3^2 \times 7$?

8 Write $2 \times 2 \times 2 \times 3$ in index form.

9 List the factors of 34.

from 3.09

Easy

Questions 10–18

10 Write $2 \times 2 \times 3 \times 3 \times 3$ in index form.

11 Which primes are factors of 12?

12 Which number is 3×7 ?

= ____ × ____

= ____

13 The lines are parallel. What is the co-interior angle to 127° ?

from 2.09

14 Write 189 as a product of primes.

15 Write $2 \times 2 \times 3 \times 3$ in index form.

16 Which primes are factors of 18?

17 Evaluate 7^2 .

from 3.07

18 Which number is $2^2 \times 5$?

Medium

Questions 19–26

19 Write 168 as a product of primes.

20 Start a factor tree with $60 = 6 \times 10$. Finish it and write 60 as a product of primes.

21 Estimate $6\,010 \div 37$.

from 3.01

22 Which number is $2^4 \times 3$?

23 Write 198 as a product of primes.

24 $\sqrt{81} + \sqrt{49}$

from 3.08

25 Start a factor tree with $36 = 9 \times 4$. Finish it and write 36 as a product of primes.

26 Which number is $2^2 \times 7$?

Medium

Questions 27–37

- 27 Start a factor tree with $24 = 4 \times 6$. Finish it and write 24 as a product of primes.

- 28 Which number is $2 \times 3^2 \times 5$?

- 29 Use a calculator: -40×36

from 1.09

- 30 Write 171 as a product of primes.

- 31 18×10

from 3.02

- 32 Write 684 as a product of primes.

- 33 Start a factor tree with $40 = 5 \times 8$. Finish it and write 40 as a product of primes.

- 34 How many factors does 19 have?

from 3.09

- 35 Which number is $3^2 \times 5^2$?

- 36 Which is larger: 2^5 or 5^2 ?

from 3.07

- 37 Write 192 as a product of primes.

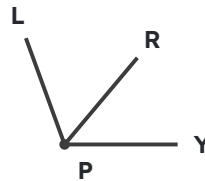
Challenging

Questions 38–47

- 38 Write 520 as a product of primes.

- 39 Use prime factors to show that 400 is a square number.

- 40 Name the three angles at P.



from 2.01

- 41 A number is $3^2 \times 5 \times 7$. Is it divisible by 2? Explain.

- 42 What number squared is 144?

from 3.08

- 43 Write 460 as a product of primes.

- 44 Use prime factors to show that 196 is a square number.

- 45 $1\,030 \div 5$

from 3.03

- 46 A number is $2 \times 3^2 \times 5$. Is it divisible by 6? Explain.

- 47 Is 96 prime or composite? Give a reason.

from 3.09

Challenging

Questions 48–57

- 48 Use prime factors to show that 225 is a square number.

- 49 A number is $2^2 \times 3 \times 7$. Is it divisible by 12? Explain.

- 50 $9 + (-7)$

from 1.04

- 51 Write 810 as a product of primes.

- 52 Find the digit sum of 2 270. Is 2 270 divisible by 3?

from 3.04

- 53 Which number is $2^2 \times 3 \times 5$?

- 54 Use prime factors to show that 144 is a square number.

- 55 Write 9×9 in index form.

from 3.07

- 56 A number is $2 \times 3^2 \times 5$. Is it divisible by 4? Explain.

- 57 $\sqrt[3]{729}$

from 3.08

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

The highest common factor (HCF) is the largest number that divides into both numbers. List the factors of each and find the largest one they share. With prime factors: multiply the primes they share.

E1 Find the HCF of 18 and 24.

E2 Find the HCF of 36 and 60 using prime factors.

E3 48 apples and 60 pears go into bags. Every bag is the same, with nothing left over. What is the largest number of bags?

Easy

Questions 1–9

1 Find the HCF of 9 and 6.

2 Is 4 a common factor of 17 and 20?

3 List the common factors of 10 and 4.

4 Find the HCF of 15 and 18.

5 Is 6 a common factor of 19 and 25?

6 List the common factors of 8 and 6.

7 Find the HCF of 8 and 6.

8 Is 6 a common factor of 10 and 30?

9 Write $2 \times 2 \times 3$ in index form.

from 3.10

Easy

Questions 10–18

10 Is 6 a common factor of 14 and 23?

11 List the common factors of 6 and 12.

12 Find the HCF of 16 and 8.

13 Which is larger: 1 or -10 ?

14 Find the HCF of 24 and 28.

24: _____

28: _____

HCF = ____

15 Is 6 a common factor of 28 and 25?

16 List the common factors of 15 and 12.

17 List the factors of 27.

18 Find the HCF of 18 and 12.

from 1.03

from 3.09

Medium

Questions 19–25

19 Find the HCF of 20 and 16.

20 Use the HCF to simplify the fraction $\frac{48}{108}$.

21 Estimate $8\,219 \div 81$.

from 3.01

22 Use prime factors to find the HCF of 63 and 28.

23 Find the HCF of 60 and 72.

24 $\sqrt{49} + \sqrt{81}$

from 3.08

25 Use the HCF to simplify the fraction $\frac{72}{108}$.

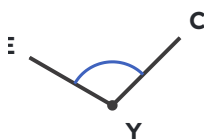
Medium

Questions 26–35

- 26 Use the HCF to simplify the fraction $\frac{12}{30}$.

- 27 Use prime factors to find the HCF of 35 and 40.

- 28 Name the marked angle with three letters.



from 2.01

- 29 Find the HCF of 78 and 65.

- 30 $68 \times 1\,000$

from 3.02

- 31 48 apples and 36 pears are shared into identical groups with none left over. What is the largest number of groups?

- 32 Use the HCF to simplify the fraction $\frac{18}{63}$.

- 33 Which primes are factors of 30?

from 3.10

- 34 Use prime factors to find the HCF of 72 and 40.

- 35 Write 34 as the sum of two primes.

from 3.09

Challenging

Questions 36–43

- 36 55 boys and 33 girls are shared into identical groups with none left over. What is the largest number of groups?

- 37 Find the HCF of 35, 14 and 21.

- 38 The highest temperature was 17°C . The lowest was -15°C . What is the difference?

from 1.10

- 39 A rectangle is 30 cm by 50 cm. It is cut into equal squares with nothing left over. What is the largest possible square?

- 40 What number squared is 16?

from 3.08

- 41 21 boys and 56 girls are shared into identical groups with none left over. What is the largest number of groups?

- 42 Find the HCF of 40, 32 and 48.

- 43 $1\,393 \div 7$

from 3.03

Challenging

Questions 44–53

- 44 Find the HCF of 18, 12 and 24.

- 45 A rectangle is 30 cm by 20 cm. It is cut into equal squares with nothing left over. What is the largest possible square?

- 46 The lines are parallel. What is the co-interior angle to 56° ?

from 2.09

- 47 30 boys and 40 girls are shared into identical groups with none left over. What is the largest number of groups?

- 48 Find the digit sum of 8 725. Is 8 725 divisible by 3?

from 3.04

- 49 Use prime factors to find the HCF of 33 and 55.

- 50 Find the HCF of 4, 6 and 8.

- 51 Is 19 prime or composite?

from 3.09

- 52 A rectangle is 22 cm by 33 cm. It is cut into equal squares with nothing left over. What is the largest possible square?

- 53 $\sqrt[3]{343}$

from 3.08

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

The lowest common multiple (LCM) is the smallest number that both numbers divide into. List the multiples of each number until you find the first one they share.

Watch your teacher.
Copy every step.

E1 Find the LCM of 4 and 6.

E2 Find the LCM of 8 and 12.

E3 Buses leave every 12 min and every 18 min. Both leave at 8:00. When do they next leave together?

Easy

Questions 1–9

1 Find the LCM of 2 and 6.

2 List the first five multiples of 7.

3 Is 12 a common multiple of 8 and 9?

4 Find the LCM of 5 and 3.

5 List the first five multiples of 9.

6 Is 20 a common multiple of 5 and 7?

7 Find the LCM of 8 and 3.

8 List the first five multiples of 8.

9 Is 3 a common factor of 15 and 9?

from 3.11

Easy

Questions 10–18

10 List the first five multiples of 6.

11 Is 24 a common multiple of 8 and 2?

12 Find the LCM of 6 and 4.

13 $-2 - (-8)$

from 1.05

14 Find the LCM of 10 and 15.

10: _____

LCM = ____

15 List the first five multiples of 3.

16 Is 51 a common multiple of 8 and 3?

17 Write $2 \times 3 \times 3$ in index form.

from 3.10

18 Find the LCM of 9 and 5.

Medium

Questions 19–26

19 Find the LCM of 10 and 5.

20 Find the LCM of 3, 4 and 6.

21 Estimate $9\ 115 \div 73$.

from 3.01

22 Find the HCF and the LCM of 13 and 14.

23 Find the LCM of 9 and 15.

24 List the primes between 70 and 90.

from 3.09

25 Find the LCM of 6, 3 and 4.

26 Find the HCF and the LCM of 18 and 11.

Medium

Questions 27–36

- 27 Find the LCM of 2, 3 and 5.
-
-
- 28 Find the HCF and the LCM of 19 and 13.
-
-
- 29 The lines are parallel. What is the co-interior angle to 62° ?
-
- from 2.09
-
- 30 Find the LCM of 12 and 15.
-
-
- 31 $37 \times 1\,000$
-
- from 3.02
- 32 Two lights flash every 4 seconds and every 12 seconds. They flash together now. When do they next flash together?
-
-
- 33 Find the LCM of 2, 5 and 6.
-
-
- 34 List the common factors of 8 and 12.
-
- from 3.11
- 35 Find the HCF and the LCM of 15 and 13.
-
-
- 36 Which number is $2^4 \times 3$?
-
- from 3.10

Challenging

Questions 37–44

- 37 Two lights flash every 6 seconds and every 12 seconds. They flash together now. When do they next flash together?
-
-
- 38 Buses leave every 14 minutes and every 6 minutes. Both leave at 9:00. When do they next leave together?
-
-
- 39 Draw an angle of 135° on the ray.
-
- 
-
- from 2.02
- 40 Hot dogs come in packs of 7 and buns in packs of 9. What is the smallest number of each you can buy to have the same number?
-
-
- 41 How many factors does 16 have?
-
-
- 42 Check that $\text{HCF} \times \text{LCM} = 11 \times 13$ for 11 and 13.
-
-
- 43 Two lights flash every 5 seconds and every 9 seconds. They flash together now. When do they next flash together?
-
-
- 44 $895 \div 5$
-
-
- from 3.03

Challenging

Questions 45–53

- 45 Buses leave every 12 minutes and every 8 minutes. Both leave at 9:00. When do they next leave together?
-
-
- 46 Hot dogs come in packs of 5 and buns in packs of 12. What is the smallest number of each you can buy to have the same number?
-
-
- 47 Which is larger: -13 or -15 ?
-
- from 1.03
- 48 Check that $\text{HCF} \times \text{LCM} = 9 \times 13$ for 9 and 13.
-
-
- 49 Find the digit sum of 7 331. Is 7 331 divisible by 3?
-
- from 3.04
- 50 Two lights flash every 6 seconds and every 7 seconds. They flash together now. When do they next flash together?
-
-
- 51 Find the HCF and the LCM of 12 and 9.
-
-
- 52 Which number is 3×7 ?
-
- from 3.10
- 53 Buses leave every 6 minutes and every 14 minutes. Both leave at 9:00. When do they next leave together?
-
-